Pyrogallol, an active compound from the medicinal plant Emblica officinalis, regulates expression of pro-inflammatory genes in bronchial epithelial...

The most relevant cause of morbidity and mortality in cystic fibrosis (CF) patients is the lung pathology characterized by chronic infection and inflammation sustained mainly by Pseudomonas aeruginosa (P. aeruginosa). Innovative pharmacological approaches to control the excessive inflammatory process in the lung of CF patients are thought to be beneficial to reduce the extensive airway tissue damage. Medicinal plants from the so-called traditional Asian medicine are attracting a growing interest because of their potential efficacy and safety. Due to the presence of different active compounds in each plant extract, understanding the effect of each component is important to pursue selective and reproducible applications. Extracts from Emblica officinalis (EO) were tested in IB3-1 CF bronchial epithelial cells exposed to the P. aeruginosa laboratory strain PAO1. EO strongly inhibited the PAO1-dependent expression of the neutrophil chemokines IL-8, GRO-alpha, GRO-gamma, of the adhesion molecule ICAM-1 and of the pro-inflammatory cytokine IL-6. Pyrogallol, one of the compounds extracted from EO, inhibited the P. aeruginosa-dependent expression of these pro-inflammatory genes similarly to the whole EO extract, whereas a second compound purified from EO, namely 5-hydroxy-isoquinoline, had no effect. These results identify Pyrogallol as an active compound responsible for the anti-inflammatory effect of EO and suggest to extend the investigation in pre-clinical studies in airway animal models in vivo, to test the efficacy and safety of this molecule in CF chronic lung inflammatory disease.